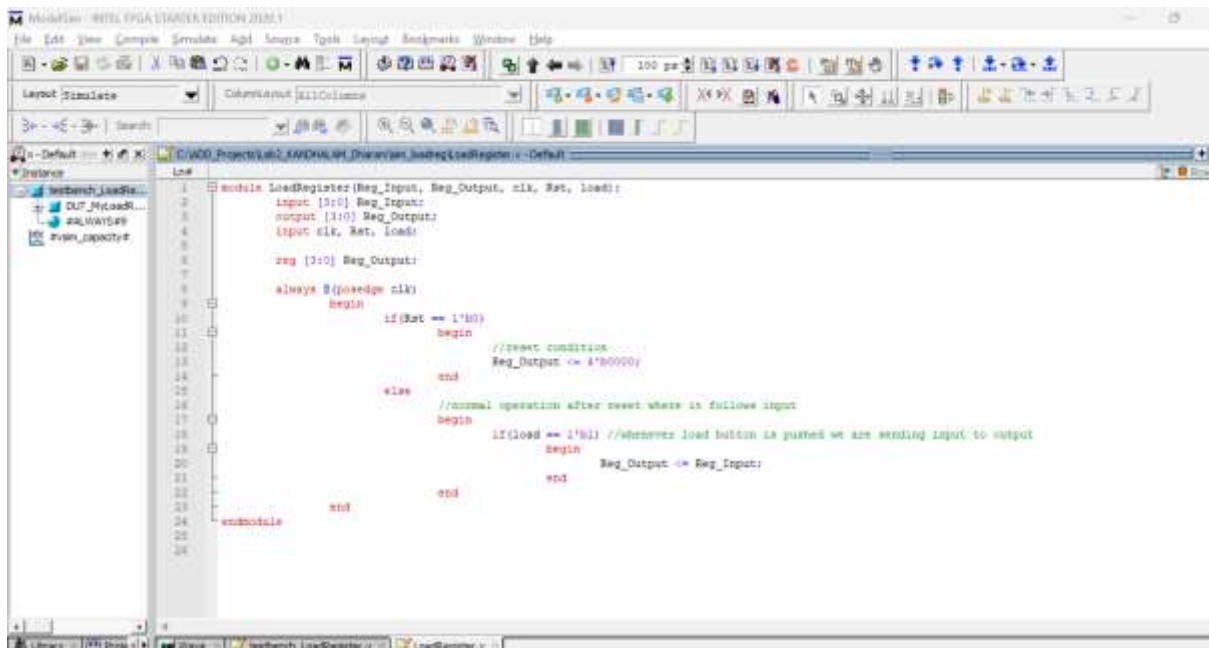


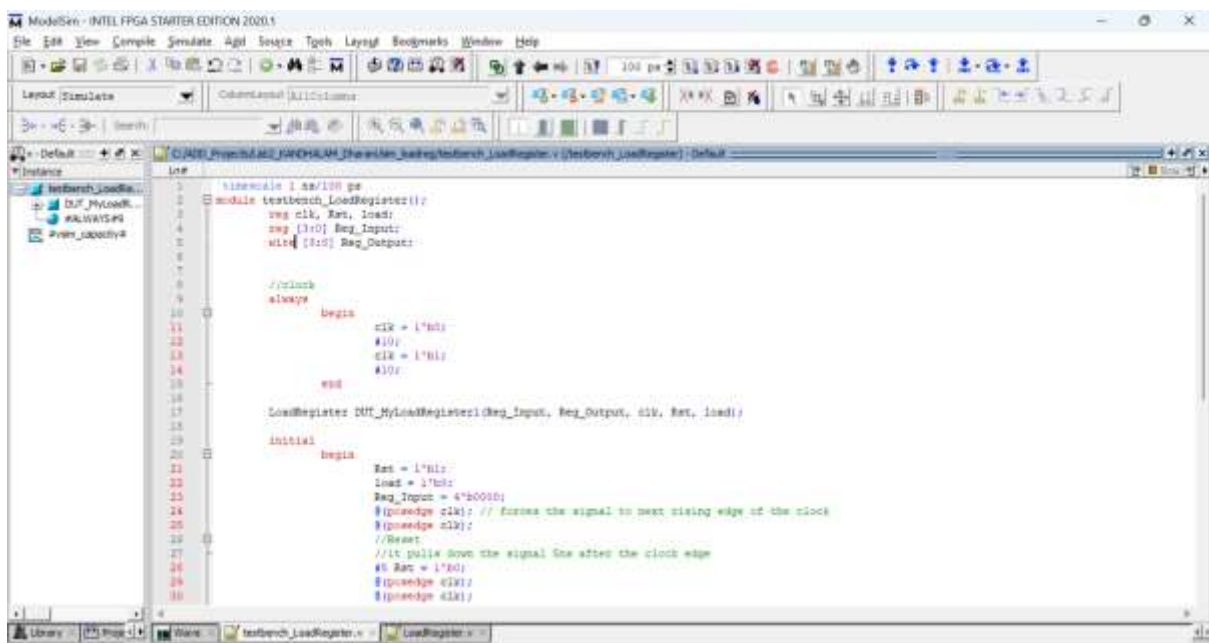
2)
Sol)

Source Code:

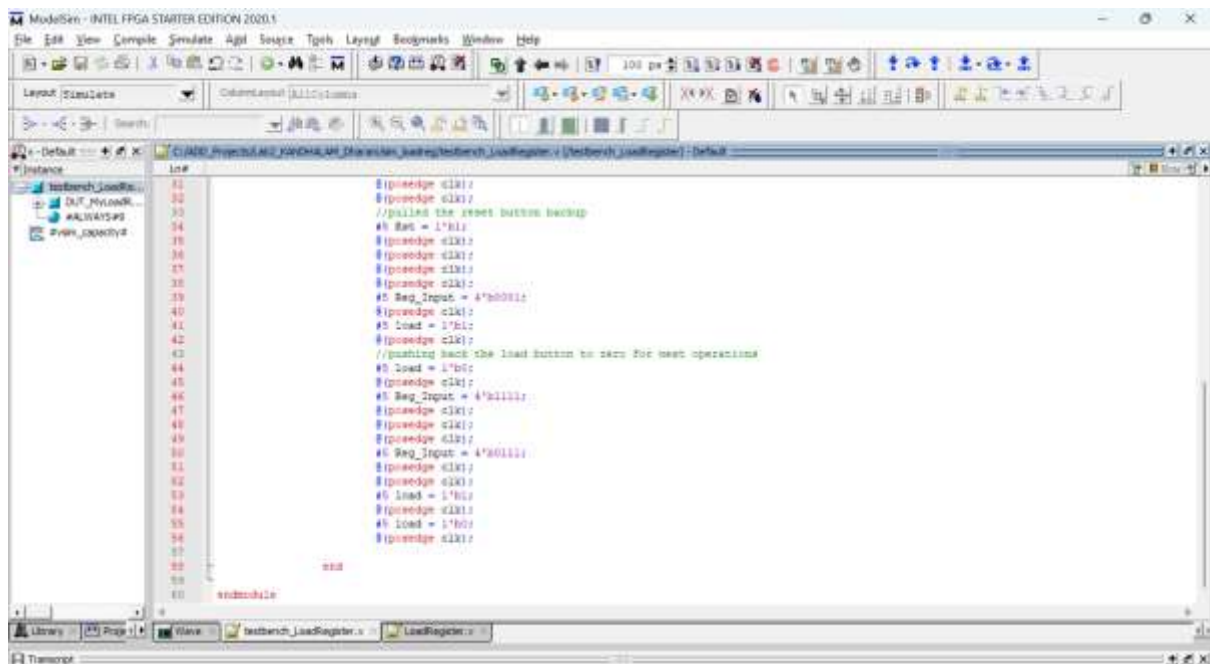
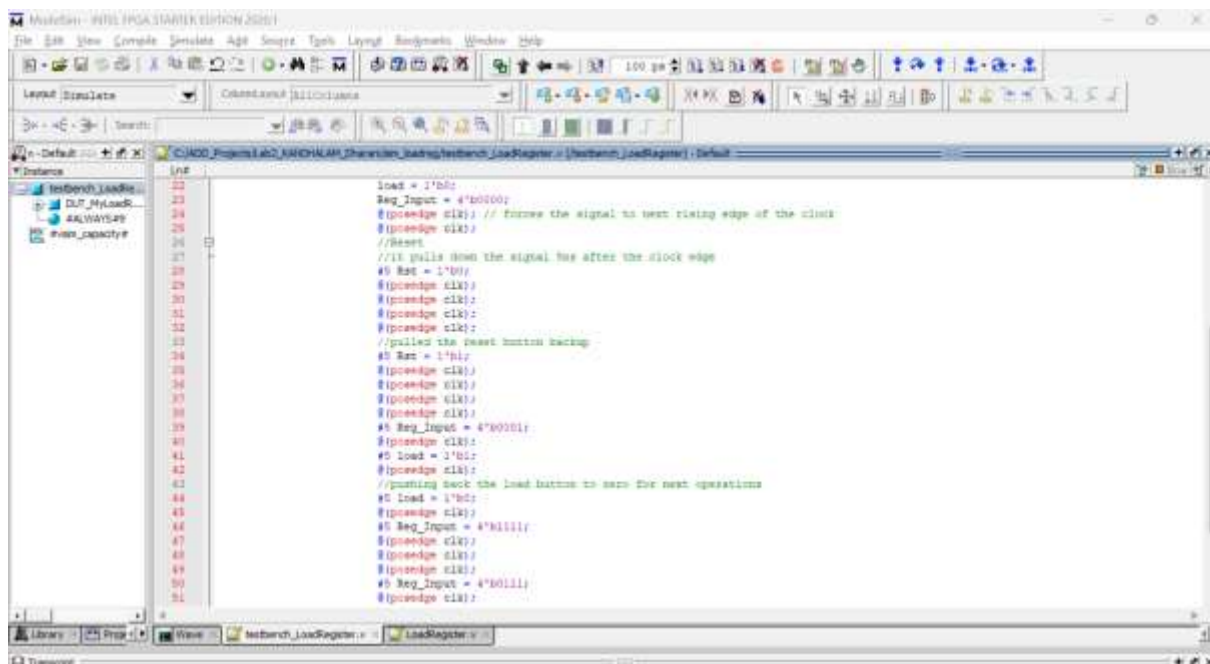


```
1 module LoadRegister(Reg_Input, Reg_Output, clk, Rst, load);
2   input [3:0] Reg_Input;
3   output [3:0] Reg_Output;
4   input clk, Rst, load;
5
6   reg [3:0] Reg_Output;
7
8   always @(posedge clk)
9   begin
10      if(Rst == 1'b0)
11      begin
12         //reset condition
13         Reg_Output <= 4'b0000;
14      end
15      else
16      //normal operation after reset where it follows input
17      begin
18         if(load == 1'b1) //whenever load button is pushed we are sending input to output
19         begin
20            Reg_Output <= Reg_Input;
21         end
22      end
23   end
24 endmodule
```

TestBench code:



```
1 timescale 1 ns/100 ps
2 module testbench_LoadRegister();
3   reg clk, Rst, load;
4   reg [3:0] Reg_Input;
5   wire [3:0] Reg_Output;
6
7   //clock
8   always
9   begin
10      clk = 1'b0;
11      #10;
12      clk = 1'b1;
13      #10;
14      clk = 1'b0;
15      #10;
16   end
17
18   LoadRegister DUT_MyLoadRegister1(Reg_Input, Reg_Output, clk, Rst, load);
19
20   initial
21   begin
22      Rst = 1'b1;
23      load = 1'b0;
24      Reg_Input = 4'b0000;
25      @(posedge clk); // forces the signal to next rising edge of the clock
26      @(posedge clk); //Reset
27      //it pulls down the signal one after the clock edge
28      #5 Rst = 1'b0;
29      @(posedge clk);
30      @(posedge clk);
```



Simulation:

Clk – as always block doesn't contain any sensitivity list, it's just keep on going 10ns high and 10ns low.

Rst – as we set reset high at beginning in initial block it is high for some time in starting of the simulation and went low.

Output – as we didn't define any out for starting simulation it is red(undefined) once reset line goes high the output went to 0000 as defined.

In this case, like reset goes to low in middle of clk transition, then the immediate next rising edge of the clk the output follows to 0000 for reset.

As there is no load button high the output doesn't changed even though the input changed to 1111.

